

1 Voting Theory

1.3 Plurality with Elimination

Example 1.3.1 Who wins the following election under the plurality method?

	3	4	4	6	2	1
1st	B	C	B	D	B	E
2nd	C	A	D	C	E	A
3rd	A	D	C	A	D	D
4th	D	B	A	E	C	B
5th	E	E	E	B	D	C

Since the winner under plurality (B) does not have the majority, with the plurality with elimination method we eliminate the last place candidate (according to plurality) and recount the votes.

Example 1.3.2 Your turn! Use the Plurality with Elimination method to determine who wins.

	44	14	70	22	80
1st	G	G	M	M	B
2nd	M	B	G	B	M
3rd	B	M	B	G	G

G is eliminated first, then M has a majority.

Example 1.3.3 Use the plurality method with elimination to determine who wins the election.

	342	214	298
1st	E	D	K
2nd	D	K	D
3rd	K	E	E

Is there a Condorcet Candidate? If so, did they win the election?

K wins under this method. But D is a Condorcet Candidate.

Example 1.3.4 Who wins the following election under the plurality with elimination method?

	37	22	12	29
1st	A	B	B	C
2nd	B	C	A	A
3rd	C	A	C	B

C is eliminated first, then A wins with the majority.

Now, suppose that after officials have announced the results, the ballots were accidentally destroyed before they could be certified, so the votes have to be recast. Ten of the people whose preference was B, C, A decide that they want to now support the winner A , so they change their votes to A, B, C , all the other votes remain the same. What are the results of the election now?

1st				
2nd				
3rd				

Definition 1.3.5 (Monotonicity Criterion) If voters change their votes to increase the preference of the winning candidate, it should not harm the candidates chances of winning.